

Chapter 4, Section 4

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3. Let the elliptic curve X be embedded in $\mathbb{P}^2$ so as to have the equation $y^2 = x(x-1)(x-\lambda)$. Show that any automorphism of X leaving $P_0 = (0 : 1 : 0)$ fixed is induced by an automorphism of $\mathbb{P}^2$ coming from the automorphism of the affine (x, y) -plane given by

$$\begin{cases} x' = ax + b \\ y' = cy. \end{cases}$$

In each of the four cases of (4.7), describe these automorphisms of $\mathbb{P}^2$ explicitly, and hence determine the structure of the group $G = \text{Aut}(X, P_0)$.

Proof. Let $f : X \rightarrow \mathbb{P}^1$ be the unique g_2^1 with $f(P_0) = \infty$, branched over $0, 1, \lambda, \infty$. If $\sigma \in G$, then by (4.4) there exists an automorphism τ of $\mathbb{P}^1$ which fixes ∞ and satisfies $f \circ \sigma = \tau \circ f$. In particular, τ sends $\{0, 1, \lambda\}$ to $\{0, 1, \lambda\}$ in some order. If $\tau = \text{id}$, then either $\sigma = \text{id}$ or σ is the unique automorphism of order 2 fixing P_0 and interchanging the sheets of f . We call it the *hyperbolic involution* of X , and we denote it by $\iota : X \rightarrow X$. It comes from the automorphism of the affine (x, y) -plane corresponding to the reflection across the x -axis:

$$\begin{cases} x' = x \\ y' = -y \end{cases}.$$

The other automorphisms are determined using the methods of (4.7). In particular, we look at the orbits of the action of S_3 on $k - \{0, 1\}$ (4.5).

$j \neq 0, 1728$ Since the orbits of λ are all distinct, $\tau = \text{id}$, and $\sigma = \text{id}$ or ι .

$j = 1728, \text{char } k \neq 3$ This is the case $\lambda = -1, \frac{1}{2}$, or 2 , and λ coincides with one other element of its orbit under the action of S_3 . For example, if $\lambda = -1$, then $\lambda = 1/\lambda$, and G is the cyclic group of order 4 generated by

$$\begin{cases} x' = -x \\ y' = iy \end{cases}.$$

$j = 0, \text{char } k \neq 3$ This is the case $\lambda = -\omega$ or $-\omega^2$, where ω is a third root of unity, and λ coincides with two other elements of its orbit under S_3 . For example, if $\lambda = -\omega$, then

$$\lambda = \frac{1}{1-\lambda} = \frac{\lambda-1}{\lambda},$$

and G is the cyclic group of order 6 generated by

$$\begin{cases} x' = \omega^2 x + 1 \\ y' = -y \end{cases}.$$

$j = 0 (= 1728), \text{char } k = 3$ If $\text{char } k = 3$ and $j = 0$, then $\lambda = -1$, and all six elements of its orbit under S_3 coincide. Note that k has no nontrivial third roots of unity, and the values $\lambda = -1, \frac{1}{2}, 2$ of the previous case coincide too. So G is the cyclic group of order 12 generated by

$$\begin{cases} x' = x + 1 \\ y' = iy \end{cases}.$$

□

4. Let X be an elliptic curve in $\mathbb{P}^2$ given by an equation of the form

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6.$$

Show that the j -invariant is a rational function of the a_i , with coefficients in $\mathbb{Q}$. In particular, if the a_i are all in some field $k_0 \subseteq k$, then $j \in k_0$ also. Furthermore, for every $\alpha \in k_0$, there exists an elliptic curve defined over k_0 with j -invariant equal to α .

Proof. Complete the square in y to get an equation of the form

$$y^2 = x^3 + Ax^2 + Bx + C,$$

with

$$A = \frac{a_1^2}{4} + a_2, \quad B = \frac{a_1a_3}{2} + a_4, \quad C = \frac{a_3^2}{4} + a_6.$$

Let α, β, γ be the distinct roots of the cubic on the right-hand side. Applying a linear transformation of x , which sends α, β to $0, 1$, transforms the equation of X into

$$y^2 = x(x-1)(x-\lambda), \quad \lambda = \frac{\gamma - \alpha}{\beta - \alpha}.$$

We can plug in λ into the j -invariant formula. Simplifying, we get a rational function of the elementary symmetric polynomials of α, β, γ (which are just A, B, C) with coefficients in $\mathbb{Q}$

$$j = 2^8 \frac{(A^2 - 3B)^3}{A^2B^2 - 4B^3 - 4A^3 + 18ABC - 27C^2}.$$

Since A, B, C are polynomials of the a_i with integer coefficients, the j -invariant is a rational function of the a_i with coefficients in $\mathbb{Q}$. $\square$

5. Let X, P_0 be an elliptic curve having an endomorphism $f : X \rightarrow X$ of degree 2.

- If we represent X as a 2-1 covering of $\mathbb{P}^1$ by a morphism $\pi : X \rightarrow \mathbb{P}^1$ ramified at P_0 , then as in (4.4), show that there is another morphism $\pi' : X \rightarrow \mathbb{P}^1$ and a morphism $g : \mathbb{P}^1 \rightarrow \mathbb{P}^1$, also of degree 2, such that $\pi \circ f = g \circ \pi'$.
- For a suitable choice of coordinates in the two copies of $\mathbb{P}^1$, show that g can be taken to be the morphism $x \mapsto x^2$.
- Now show that g is branched over two of the branch points of π , and that g^{-1} of the other two branch points of π consists of the four branch points of π' . Deduce a relation involving the invariant λ of X .
- Solving the above, show that there are just three values of j corresponding to elliptic curves with an endomorphism of degree 2, and find the corresponding values of λ and j .

Proof.

- The morphism $\pi \circ f$ corresponds to the linear system $|f^*(2P_0)|$. Notice that we can write $f^*(2P_0) = 2f^*(P_0)$, which gives a hint as to how to factor $\pi \circ f$. In particular, we can factor $\pi \circ f$ through the morphism $\pi' : X \rightarrow \mathbb{P}^1$ given by the linear system $|f^*(P_0)|$ on X , followed by the morphism $g : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ given by the linear system $|\pi'_*(f^*P_0)|$ on $\mathbb{P}^1$.
- Any degree two morphism $g : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ is branched over two points, and we can choose coordinates in the two copies of $\mathbb{P}^1$ so that these two points are $0, \infty$ in the target and source. Then g is given by the morphism $x \mapsto x^2$. Indeed, on some local affine patch, g is given by a degree two polynomial in x , say $g(x) = (x - \alpha)(x - \beta)$. Then g is ramified at $x = \frac{1}{2}(\alpha + \beta)$ and ∞ , and we can choose coordinates so that these two points are $0, \infty$. Similarly, on the target, we can choose coordinates so that the two branch points of g are $0, \infty$. Then g is given by a degree two polynomial with roots at $0, \infty$, which is just $x \mapsto x^2$.
- The morphism $\pi \circ f$ is generically four-to-one, and it is branched over the branch points of π because any morphism of elliptic curves is unramified by the Riemann-Hurwitz formula. This just says $(\pi \circ f)^{-1}$ of a branch point of π consists of two points. Since g is a morphism of degree two ramified at exactly two points, and π' is ramified at exactly four points, g must be branched over two of the branch points of π ,

and g^{-1} of the other two branch points of π consists of the four branch points of π' . Note that one of the points g is branched over is given by

$$g(\pi'(f^{-1}P_0)) = \pi(f(f^{-1}P_0)) = \pi(P_0).$$

Now, choose coordinates in the two copies of $\mathbb{P}^1$ so that the branch points of g are $0, \infty$, the other two branch points of π are $1, \lambda$, and g is given by $x \mapsto x^2$. Then the four branch points of π' are $\pm 1, \pm \sqrt{\lambda}$. Consider the linear transformation of $\mathbb{P}^1$ sending $-1, 1, \sqrt{\lambda}$ to $0, 1, \infty$, respectively, which is given by

$$\phi(x) = \frac{(1 - \sqrt{\lambda})(x + 1)}{2(x - \sqrt{\lambda})}$$

Then the branch points of π' are sent to $0, 1, \infty, \phi(-\sqrt{\lambda})$, where

$$\lambda' := \phi(-\sqrt{\lambda}) = -\frac{(1 - \sqrt{\lambda})^2}{4\sqrt{\lambda}}.$$

Since λ' and λ give the same j -invariant, they must be related by a fractional linear transformation in S_3 . In particular, λ' is in the orbit of λ under the action of S_3 on $k - \{0, 1\}$ (4.5).

(d) Solving for λ in the above, we get the following three unique values of λ modulo the action of S_3 :

$$\lambda = 2, \quad 3 + 2\sqrt{2}, \quad \frac{1 + 3i\sqrt{7}}{2}.$$

The corresponding values of j are

$$j = 2^6 \cdot 3^3, \quad 2^6 \cdot 5^3, \quad -3^3 \cdot 5^3.$$

□

7. The Dual of a Morphism. Let X and X' be elliptic curves over k , with base points P_0, P'_0 .

- (a) If $f : X \rightarrow X'$ is any morphism, use (4.11) to show that $f^* : \text{Pic } X' \rightarrow \text{Pic } X$ induces a homomorphism $\hat{f} : (X', P'_0) \rightarrow (X, P_0)$. We call this the *dual* of f .
- (b) If $f : X \rightarrow X'$ and $g : X' \rightarrow X''$ are two morphisms, then $\widehat{g \circ f} = \hat{g} \circ \hat{f}$.
- (c) Assume $f(P_0) = P'_0$, and let $n = \deg f$. Show that if $Q \in X$ is any point, and $f(Q) = Q'$, then $\hat{f}(Q') = n_X(Q)$. Conclude that $f \circ \hat{f} = n_{X'}$ and $\hat{f} \circ f = n_X$.
- (d) If $f, g : X \rightarrow X'$ are two morphisms preserving the base points P_0, P'_0 , then $\widehat{f + g} = \hat{f} + \hat{g}$.
- (e) Using (d), show that for any $n \in \mathbb{Z}$, $\hat{n}_X = n_X$. Conclude that $\deg n_X = n^2$.
- (f) Show for any f that $\deg \hat{f} = \deg f$.

Proof. Let the pair $(X, \mathcal{P})$ (resp. $(X', \mathcal{P}')$) be the Jacobian of (X, P_0) (resp. (X', P'_0)). If $\mathcal{C}$ is any category, and T is any object in $\mathcal{C}$, then denote by $h_T : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ the contravariant Hom functor defined by $U \mapsto \text{Hom}_{\mathcal{C}}(U, T)$. Recall that X and $\mathcal{P} \in \text{Pic}^\circ(X/X)$ satisfy the following universal property: if T is any scheme of finite type over k , and for any $\mathcal{M} \in \text{Pic}^\circ(X/T)$, there is a unique morphism $f : T \rightarrow X$ such that $f^*\mathcal{P} \cong \mathcal{M}$ in $\text{Pic}^\circ(X/T)$. In other words, $(X, \mathcal{P})$ represents the functor $\text{Pic}^\circ(X/-)$. Lastly, $(X, \mathcal{P})$ is a group object in the category of schemes of finite type over k , so for any T , $h_X(T)$ has a natural group structure induced by $\text{Pic}^\circ(X/T)$.

- (a) Any morphism $f : X \rightarrow X'$ defines a natural transformation $f^* : \text{Pic}^\circ(X'/-) \rightarrow \text{Pic}^\circ(X/-)$ of group functors in the following way (4.10.2). If T is any scheme of finite type over k , then $f' = f \times 1_T : X' \times T \rightarrow X \times T$ induces a group homomorphism $f^* : \text{Pic}(X' \times T) \rightarrow \text{Pic}(X \times T)$. Let $p : X \times T \rightarrow T$ and $p' : X' \times T \rightarrow T$ be the projection maps. The image of f^* lies in $\text{Pic}^\circ(X \times T)$, and $p = p' \circ f'$ implies $f^*p'^*\text{Pic}(T) = p^*\text{Pic}(T)$. Hence, we have a well-defined group homomorphism $f_T^* : \text{Pic}^\circ(X'/T) \rightarrow \text{Pic}^\circ(X/T)$ for each scheme T of finite type over k .

It remains to show f^* is functorial. Let $g : S \rightarrow T$ be any morphism of schemes of finite type over k . We can apply $\text{Pic}(-)$ to the following commutative diagram

$$\begin{array}{ccc} X \times S & \xrightarrow{f \times 1_S} & X' \times S \\ 1_X \times g \downarrow & & \downarrow 1_{X'} \times g \\ X \times T & \xrightarrow{f \times 1_T} & X' \times T \end{array}$$

which gives a commutative diagram of Picard groups with arrows given by taking inverse images of invertible sheaves.

Now, we identify h_X and $h_{X'}$ with $\text{Pic}^\circ(X/-)$ and $\text{Pic}^\circ(X'/-)$, respectively. Yoneda's lemma says there exists a natural bijection between the set of morphisms $X' \rightarrow X$ and the set of natural transformations of functors $h_{X'} \rightarrow h_X$. Hence, there exists a unique morphism $\hat{f} \in \text{Hom}(X', X)$ such that $f^* \in \text{Hom}(h_{X'}, h_X)$ is given by composition with $\hat{f}$. The invertible sheaves $\mathcal{O}_X$ and $\mathcal{O}_{X'}$ correspond to $P_0 \in X$ and $P'_0 \in X'$, respectively, and structure sheaves pullback, so we conclude that $\hat{f}(P'_0) = P_0$.

- (b) This is a consequence of the uniqueness property of $\hat{f}$ in $\text{Hom}(X', X)$, and the analogous fact for inverse image of sheaves, i.e., $(g \circ f)^* = f^* \circ g^*$.
- (c) The points $Q' \in X'$ and $\hat{f}(Q') \in X$ correspond to $\mathcal{O}_{X'}(Q' - P'_0)$ in $\text{Pic}^\circ(X'/k)$ and $f^*\mathcal{O}_{X'}(Q' - P'_0)$ in $\text{Pic}^\circ(X/k)$, respectively. Thus, it is enough to show the linear equivalence of the following divisors

$$f^*(Q' - P'_0) \sim n_X(Q) - P_0.$$

By the hypothesis, $Q' - P'_0 = f_*(Q - P_0)$, which by (Ex. 2.6) implies

$$f^*(Q' - P'_0) = n(Q - P_0).$$

The right-hand side is linearly equivalent to $n_X(Q) - P_0$ because the group law on X is induced by that of $\text{Pic}^\circ(X/k)$, so we conclude that $\hat{f} \circ f = n_X$. For the second corollary, observe that f and multiplication by n commutes since f is a morphism of proper group schemes. Hence, we have

$$(f \circ \hat{f})(Q') = f(n_X(Q)) = n_{X'}(f(Q)) = n_{X'}(Q').$$

- (d) It is enough to show for any $\mathcal{L}' \in \text{Pic}^\circ(X')$, that

$$(f + g)^*\mathcal{L}' \cong f^*\mathcal{L}' \otimes g^*\mathcal{L}'.$$

Let Q' be the closed point in X' such that $\mathcal{L}' \cong \mathcal{O}_{X'}(Q' - P'_0)$. There exists a bijection

$$\begin{aligned} \text{Hom}_{\mathbf{Sch}(k)}(X, X') &\xrightarrow{\sim} \text{Pic}^\circ(X'/X) \\ f : X \rightarrow X' &\mapsto \mathcal{L}_f := (f \times 1_{X'})^*\mathcal{P}', \end{aligned}$$

where the group law on the left-hand side is induced by $\text{Pic}^\circ(X'/X)$, i.e., $\mathcal{L}_{f+g} = \mathcal{L}_f \otimes \mathcal{L}_g$ is a representative of $f+g$ in $\text{Pic}(X \times X')$ for any $f, g : X \rightarrow X'$. Let $\sigma : X \rightarrow X \times X'$ be the section $x \mapsto (x, Q')$. If we assume $\mathcal{P}' = \mathcal{O}_{X' \times X'}(\Delta - X' \times P'_0)$ as in (4.11), then $\sigma^*\mathcal{L}_f \cong f^*\mathcal{O}_{X'}(Q')$, so replacing $\mathcal{P}'$ by $\mathcal{P}' \otimes p_2^*\mathcal{O}_{X'}(-P_0)$, we get $\sigma^*\mathcal{L}_f \cong f^*\mathcal{L}'$. Hence,

$$(f + g)^*\mathcal{L}' \cong \sigma^*\mathcal{L}_{f+g} \cong \sigma^*\mathcal{L}_f \otimes \sigma^*\mathcal{L}_g \cong f^*\mathcal{L}' \otimes g^*\mathcal{L}'.$$

- (e) Proceeding by induction on n reduces the statement to the case $n = 2$. The dual of the identity is the identity. Hence, $\hat{2}_X = \widehat{1_X + 1_X} = 1_X + 1_X = 2_X$ by (d). According to (c), $\hat{n}_X \circ n_X$ is multiplication by $\deg n_X$. Since $\hat{n}_X \circ n_X = n_X^2$, we conclude that $\deg n_X = n^2$.
- (f) By (c) and (e), $\deg \hat{f} \circ f = n^2$. The degree of morphisms of curves is multiplicative with respect to composition. Hence, $\deg \hat{f} = n$.

□

9. We say two elliptic curves X, X' are *isogenous* if there is a finite morphism $f : X \rightarrow X'$.

- (a) Show that isogeny is an equivalence relation.

- (b) For any elliptic curve X , show that the set of elliptic curves X' isogenous to X , up to isomorphism is countable.

Proof.

- (a) The identity morphism is trivially finite, so any elliptic curve is isogenous to itself. A composition of finite morphisms is finite, so isogeny satisfies transitivity. Lastly, (Ex. 4.7) implies isogeny is reflexive. Hence, isogeny is an equivalence relation.
- (b) Since $\ker f$ is a closed subgroup scheme of X , it is a finite set of points, all of which are torsion under the group law of X . Every torsion point of X lies in the kernel of a multiplication map $n_X : X \rightarrow X$ for some $n > 0$. By (Ex. 4.7), n_X is an étale covering of degree n^2 , which implies X has exactly n^2 points of order n . Hence, there are countably many torsion points on X . The set of finite subsets of a countable set is countable, so there are only countably many candidates for $\ker f$. Since X' is uniquely determined by X and $\ker f$, the set of elliptic curves isogenous to X is countable. □

11. Let X be an elliptic curve over $\mathbb{C}$, defined by the elliptic functions with periods $1, \tau$. Let R be the ring of endomorphisms of X .

- (a) If $f \in R$ is a nonzero endomorphism corresponding to complex multiplication by α as in (4.18), show that $\deg f = |\alpha|^2$.
- (b) If $f \in R$ corresponds to $\alpha \in \mathbb{C}$ again, show that the dual $\hat{f}$ of (Ex. 4.7) corresponds to the complex conjugate $\bar{\alpha}$ of α .
- (c) If $\tau \in \mathbb{Q}(\sqrt{-d})$ happens to be integral over $\mathbb{Z}$, show that $R = \mathbb{Z}[\tau]$.

Proof.

- (a) It is not hard to see $\deg f$ is the size of the quotient $|\Lambda/\alpha\Lambda|$. Equivalently, it is the ratio between the areas of the parallelograms with vertices $\alpha, \alpha\tau$, and $1, \tau$. But α just scales the area by $|\alpha|^2$. Hence, $\deg f = |\alpha|^2$.
- (b) By (Ex. 7c), $\hat{f} \circ f$ is multiplication by $|\alpha|^2$, but f is multiplication by α , and $\bar{\alpha}$ is the unique element in $\mathbb{C}$ such that $\alpha\bar{\alpha} = |\alpha|^2$. Hence, $\hat{f}$ is multiplication by $\bar{\alpha}$.
- (c) The ring of integers of $\mathbb{Q}(\sqrt{-d})$ is given by

$$\begin{cases} \mathbb{Z}[\sqrt{-d}] & d \equiv 1, 2 \pmod{4} \\ \mathbb{Z}\left[\frac{1 + \sqrt{-d}}{2}\right] & d \equiv 3 \pmod{4} \end{cases}.$$

By (4.19), if $\tau = r + s\sqrt{-d}$ for some $r, s \in \mathbb{Q}$, the endomorphism ring R is

$$R = \{a + b\tau \mid a, b \in \mathbb{Z}, \quad 2br, b(r^2 + ds^2) \in \mathbb{Z}\}.$$

Suppose τ is integral over $\mathbb{Z}$. If $d \equiv 1, 2 \pmod{4}$, then $r, s \in \mathbb{Z}$, so $a + b\tau \in R$ for all $a, b \in \mathbb{Z}$. If $d \equiv 3 \pmod{4}$, then we can write $\tau = (r' + s'\sqrt{-d})/2$ with $r', s' \in \mathbb{Z}$ such that $r' \equiv s' \pmod{2}$. For any $b \in \mathbb{Z}$, the first condition $2br = br' \in \mathbb{Z}$ is easily satisfied. Also, we have

$$r^2 + ds^2 = \frac{r'^2 + ds'^2}{4} \in \mathbb{Z}$$

for all $b \in \mathbb{Z}$ since $r'^2, s'^2 \equiv 0, 1 \pmod{4}$. Hence, $R = \mathbb{Z}[\tau]$. □

12. Again let X be an elliptic curve over $\mathbb{C}$ determined by the elliptic functions with periods $1, \tau$, and assume that τ lies in the region G of (4.15B).

- (a) If X has any automorphisms leaving P_0 fixed other than ± 1 , show that either $\tau = i$ or $\tau = \omega$, as in (4.20.1) and (4.20.2). This gives another proof of the fact (4.7) that there are only two curves, up to isomorphism, having automorphisms other than ± 1 .

- (b) Now show that there are exactly three values of τ for which X admits an endomorphism of degree 2. Can you match these with the three values of j determined in (Ex. 4.5)?

Proof.

- (a) By (Ex. 4.11), τ must be a root of unity in G . The only roots of unity in G are i and ω .
(b) By (Ex. 4.11), we need to find the values of τ such that Λ contains an element of norm 2. In particular, if we write $m + n\tau \in \Lambda$, then we need to determine when

$$m^2 + n^2|\tau|^2 + 2mn \operatorname{Re}(\tau) = 4$$

admits a solution for some $m, n \in \mathbb{Z}$. Using the bounds on $|\tau|$ and $\operatorname{Re}(\tau)$ as specified in (4.15B), the only possible values of τ are

$$\tau = i, \quad i\sqrt{2}, \quad -\frac{1}{2} + i\frac{\sqrt{7}}{2}.$$

The corresponding values of j are

$$j = 2^6 \cdot 3^3, \quad 2^6 \cdot 5^3, \quad -3^3 \cdot 5^3.$$

□

- 13.** If $p = 13$, there is just one value of j for which the Hasse invariant of the corresponding curve is 0. Find it.

Proof. The polynomial $h_p(\lambda)$ of (4.22) for $p = 13$ is

$$h_{13}(\lambda) = \lambda^6 + 10\lambda^5 + 4\lambda^4 + 10\lambda^3 + 4\lambda^2 + 10\lambda + 1,$$

and an elliptic curve has Hasse invariant 0 if and only if $h_{13}(\lambda) = 0$. On the otherhand, consider the j -invariant

$$j = 2^8 \frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2(\lambda - 1)^2}.$$

Expanding above modulo 13 gives

$$\lambda^6 + 10\lambda^5 + (10j + 6)\lambda^4 + (6j + 6)\lambda^3 + (10j + 6)\lambda^2 + 10\lambda + 1 = 0.$$

Matching coefficients with $h_{13}(\lambda)$ and solving for j gives $j \equiv 5 \pmod{13}$.

□

- 14.** The Fermat curve $X : x^3 + y^3 = z^3$ gives a nonsingular curve in characteristic p for every $p \neq 3$. Determine the set $\mathfrak{P} = \{p \neq 3 \mid X_{(p)} \text{ has Hasse invariant } 0\}$, and observe (modulo Dirichlet's theorem) that it is a set of primes of density $1/2$.

Proof. Using (4.21) with $f(x, y, z) = x^3 + y^3 - z^3$, we find that the coefficient of $(xyz)^{p-1}$ in f^{p-1} is zero if and only if $p \equiv 2 \pmod{3}$. By Dirichlet's theorem, the density of $\mathfrak{P}$ is $1/\phi(3) = 1/2$. □

- 15.** Let X be an elliptic curve over a field k of characteristic p . Let $F' : X_p \rightarrow X$ be the k -linear Frobenius morphism (2.4.1). Use (4.10.7) to show that the dual morphism $\hat{F}' : X \rightarrow X_p$ is separable if and only if the Hasse invariant of X is 1. Now use (Ex. 4.7) to show that if the Hasse invariant is 1, then the subgroup of points of order p on X is isomorphic to $\mathbb{Z}/p$; if the Hasse invariant is 0, then it is 0.

Proof. By (III, Ex. 8.1) and (4.10.7), if $f : X \rightarrow X'$ is any morphism of elliptic curves over k , then the induced map of cohomology $f^* : H^1(X, \mathcal{O}_{X'}) \rightarrow H^1(X, \mathcal{O}_X)$ is precisely the map of tangent spaces at 0 of the dual map $\hat{f} : X' \rightarrow X$. By the Riemann-Hurwitz formula, any separable morphism of elliptic curves is not ramified. Thus, a morphism of elliptic curves $f : X \rightarrow X'$ is inseparable if and only if f is ramified at a point, in which case f is ramified everywhere by the group structure of elliptic curves. Also, any morphism of curves is ramified at a point if and only if the map of tangent spaces is not an isomorphism. Hence, $\hat{F}'$ is separable if and only if the map of tangent spaces $F'^* : H^1(X, \mathcal{O}_X) \rightarrow H^1(X_p, \mathcal{O}_{X_p})$ is not the zero map if and only if the Hasse invariant of X is 1.

The subgroup of points of order p on X is the kernel of the multiplication by p morphism $p_X : X \rightarrow X$. Since the k -linear Frobenius morphism F' has degree p , we have $p_X = F' \circ \hat{F}'$ by (Ex. 4.7c). Also, F' is a homeomorphism on the underlying topological spaces, i.e., F' is a one-to-one map. If the Hasse invariant of X is 1, then $\hat{F}'$ is a separable morphism of degree p , so $\hat{F}'$ is a p -to-one map. Hence, p_X is a p -to-one map, and $(\ker p_X)[k] \cong \mathbb{Z}/p$. If the Hasse invariant of X is 0, then $p_X = F' \circ \hat{F}'$ is a purely inseparable morphism of curves. Hence, p_X is a one-to-one map, and $(\ker p_X)[k] = 0$. $\square$

16. Again let X be an elliptic curve over k of characteristic p , and suppose X is defined over the field $\mathbb{F}_q$ of $q = p^r$ elements, i.e., $X \subseteq \mathbb{P}^2$ can be defined by an equation with coefficients in $\mathbb{F}_q$. Assume also that X has a rational point over $\mathbb{F}_q$. Let $F' : X_q \rightarrow X$ be the k -linear Frobenius with respect to q .

- (a) Show that $X_q \cong X$ as schemes over k , and that under this identification, $F' : X_q \rightarrow X$ is the map obtained by the q th-power map on the coordinates of points of X , embedded in $\mathbb{P}^2$.
- (b) Show that $1_X - F'$ is a separable morphism and its kernel is just the set $X(\mathbb{F}_q)$ of points of X with coordinates in $\mathbb{F}_q$.
- (c) Using (Ex. 4.7), show that $F' + \hat{F}' = a_X$ for some integer a , and that $N = q - a + 1$, where $N = \#X(\mathbb{F}_q)$.
- (d) Use the fact that $\deg(m + nF') > 0$ for all $m, n \in \mathbb{Z}$ to show that $|a| \leq 2\sqrt{q}$. This is Hasse's proof of the analogue of the Riemann hypothesis for elliptic curves.
- (e) Now assume $q = p$, and show that the Hasse invariant of X is 0 if and only if $a \equiv 0 \pmod{p}$. Conclude that for $p \geq 5$ that X has Hasse invariant 0 if and only if $N = p + 1$.

Proof.

- (a) Suppose X is locally isomorphic to $\text{Spec } A$, where $A = k[x, y]/(f)$ and $f(x, y)$ is some degree three polynomial with coefficients in $\mathbb{F}_q \subset k$. Then X_p is locally isomorphic to $\text{Spec}(k \otimes_q A)$, where the copy of k here has the q th-power Frobenius for the structure morphism as a k -algebra. The image of f under the identification

$$\begin{aligned} k \otimes_q k[x, y] &\cong k[x, y] \\ (b \cdot 1)x &\mapsto b^q x, \end{aligned}$$

gives a defining equation $f^{(q)}$ for X_q . Notice that $f^{(q)}$ is obtained from f by raising each coefficient of f to the q th-power. Since each coefficient of f is in $\mathbb{F}_q$, we see that $f^{(q)}$ and f are actually equal in $k[x, y]$. Hence, $X_q \cong X$ as schemes over k .

Under this identification, $F' : X_q \rightarrow X$ is obtained from the restriction of the Frobenius on $\mathbb{P}^2$. The k -linear Frobenius morphism on an affine coordinate ring of $\mathbb{P}^2$ can be written as

$$\begin{aligned} F'^* : k[x, y] &\rightarrow k[x, y] \\ g(x, y) &\mapsto g(x^q, y^q), \end{aligned}$$

which is exactly the ring homomorphism that gives the q th-power map on the coordinates.

- (b) If $t \in \mathcal{O}_{P, X}$ is a local parameter of X , then $(1_X - F')^* t = t - t^p$. Since $d(t - t^p) = dt$, the map $1_X - F'$ is ramified nowhere. Hence, $1_X - F'$ is a separable morphism. Under the embedding $X \subset \mathbb{P}^2$, the kernel of $1_X - F'$ consists of the points fixed by the map on X obtained by the q th-power map on the coordinates of points of X . Since $\mathbb{F}_q$ is precisely the fixed field of k under the q th-power Frobenius, the kernel of $1_X - F'$ consists of the points of $X \subset \mathbb{P}^2$ with coordinates in $\mathbb{F}_q$.
- (c) Recall from (Ex. 4.7), (Ex. 4.15), and (Ex. 4.16b) that

$$(1_X - F') \circ \widehat{(1_X - F')} = N_X, \quad F' \circ \hat{F}' = q_X.$$

Expanding the left-hand side of the first identity gives

$$(1_X - F') \circ \widehat{(1_X - F')} = 1_X - (F' + \hat{F}') + q_X.$$

Hence, we have

$$F' + \hat{F}' = q_X + 1_X - N_X.$$

(d) For all $m, n \in \mathbb{Z}$, we have the inequality

$$m^2 + n^2q - mna = \deg(m - nF') > 0.$$

Rearranging gives

$$|a| < \left| \frac{m}{n} + \frac{n}{m}q \right|$$

The real-valued function $f(t) = t + q/t$ has a local extremas at $(\sqrt{q}, \pm 2\sqrt{q})$. The rational numbers are dense in the real numbers, so we conclude that $|a| \leq 2\sqrt{q}$. I wonder if it is every the case $a = 2\sqrt{q}$ when $\sqrt{q}$ is a perfect square.

(e) By (Ex. 4.15), the Hasse invariant of X is 0 if and only if $\hat{F}'$ is inseparable. If $t \in \mathcal{O}_{P,X}$ is a local parameter, then

$$(F' + \hat{F}')^*t = t^p + \hat{F}'^*t,$$

so $\hat{F}'$ is inseparable if and only if $F' + \hat{F}'$ is inseparable. The conclusion follows from the fact that a_X is inseparable if and only if $a \equiv 0 \pmod{p}$. For $p \geq 5$, $a \equiv 0 \pmod{p}$ and $|a| \leq 2\sqrt{p}$ can be both satisfied if and only if $a = 0$. What about the cases for $p < 5$?

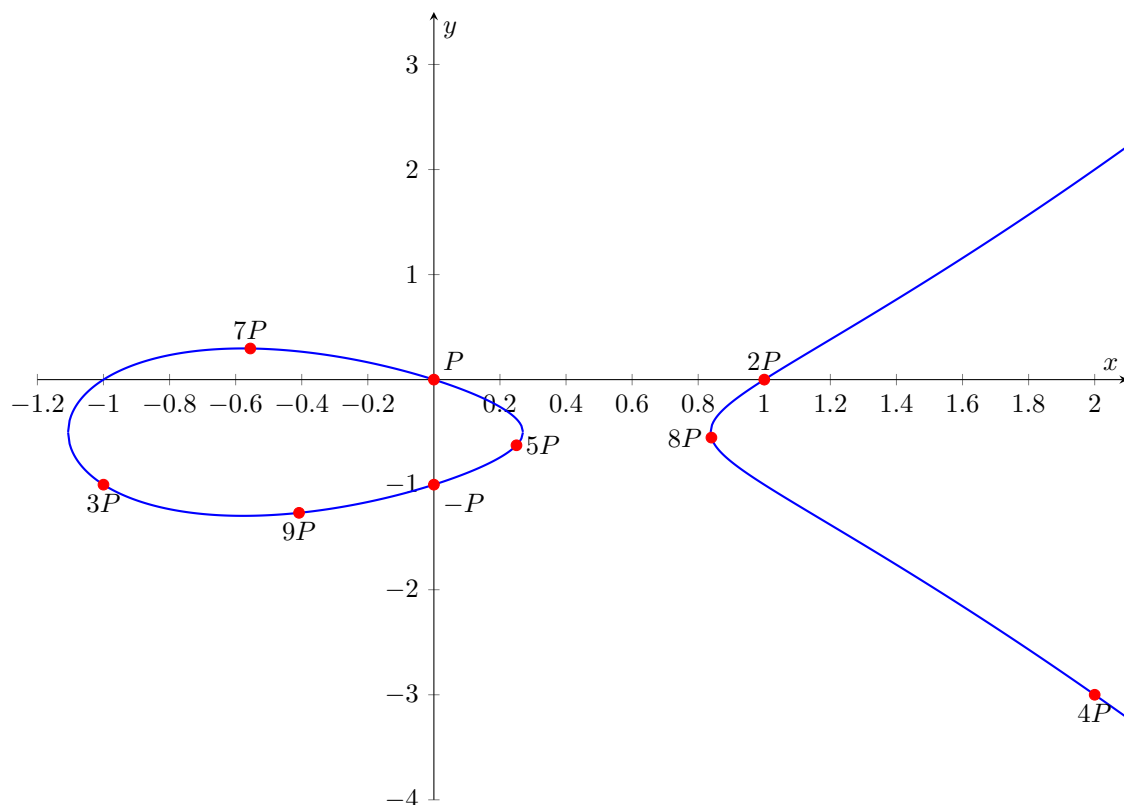
□

17. Let X be the curve $y^2 + y = x^3 - x$ of (4.23.8).

- (a) If $Q = (a, b)$ is a point on the curve, compute the coordinates of the point $P + Q$, where $P = (0, 0)$, as a function of a, b . Use this formula to find the coordinates nP , $n = 1, 2, \dots, 10$.
- (b) This equation defines a nonsingular curve over $\mathbb{F}_q$ for all $p \neq 37$.

Proof.

$$(a) \quad P + Q = \left(-\frac{a+b}{a^2}, \frac{b(a+b)}{a^3} - 1 \right), \quad -P = (0, -1), \quad 2P = (1, 0)$$



(b) Let X be a cubic curve in $\mathbb{P}^2$ given by an equation of the form

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \quad a_i \in k.$$

Define the following quantities

$$\begin{aligned} b_2 &= a_1^2 + 4a_2 \\ b_4 &= 2a_4 + a_1a_3 \\ b_6 &= a_3^2 + 4a_6 \\ b_8 &= a_1^2a_6 + 4a_2a_6 - a_1a_3a_4 + a_2a_3^2 - a_4^2. \end{aligned}$$

The discriminant of X is given by

$$\Delta = -b_2^2b_8 - 8b_4^3 - 27b_6^2 + 9b_2b_4b_6,$$

and X defines a nonsingular curve over $\mathbb{F}_q$ if and only if $\Delta \neq 0$ in $\mathbb{F}_q$. For our elliptic curve, the quantities b_i are

$$b_2 = 0, \quad b_4 = -2, \quad b_6 = 1, \quad b_8 = -1,$$

so the determinant of X is $\Delta = 37$. Hence, X defines a nonsingular curve over $\mathbb{F}_q$ for all $p \neq 37$. □

20. Let X be an elliptic curve over a field k of characteristic $p > 0$, and let $R = \text{End}(X, P_0)$ be its ring of endomorphisms.

- (a) Let X_p be the curve over k defined by changing the k -structure of X (2.4.1). Show that $j(X_p) = j(X)^{1/p}$. Thus, $X \cong X_p$ over k if and only if $j \in \mathbb{F}_p$.
- (b) Show that p_X in R factors into a product of $\pi\hat{\pi}$ of two elements of degree p if and only if $X \cong X_p$. In this case, the Hasse invariant of X is 0 if and only if π and $\hat{\pi}$ are associates in R .
- (c) If $\text{Hasse}(X) = 0$ show in any case $j \in \mathbb{F}_{p^2}$.
- (d) For any $f \in R$, there is an induced map $f^* : H^1(\mathcal{O}_X) \rightarrow H^1(\mathcal{O}_X)$. This must be multiplication by an element $\lambda_f \in k$. So we obtain a ring homomorphism $\varphi : R \rightarrow k$ by sending f to λ_f . Show that any $f \in R$ commutes with the (nonlinear) Frobenius morphism $F : X \rightarrow X$, and conclude that if $\text{Hasse}(X) \neq 0$, then the image of φ is $\mathbb{F}_p$. Therefore, R contains a prime ideal $\mathfrak{p}$ with $R/\mathfrak{p} \cong \mathbb{F}_p$.

Proof.

- (a) Some of the statements proved in (Ex. 4.16) hold here as well. In particular, under the identification of the k -linear Frobenius morphism $F^* : k[x, y] \rightarrow k[x, y]$ defined by $x, y \mapsto x^p, y^q$, let $f(x, y)$ be a cubic polynomial that locally defines X in $\mathbb{P}^2$. The image of $f(x, y)$ under F^* is $f(x^p, y^p) = 0$, so if we write

$$f(x, y) = y^2 - x(x-1)(x-\lambda),$$

then

$$F^*f(x, y) = y^{2p} - x^p(x^p-1)(x^p-\lambda).$$

Since k is algebraically closed, there exists a unique p th root $\lambda^{1/p}$ of λ . Thus, we can factorize

$$y^{2p} - x^p(x^p-1)(x^p-\lambda) = (y^2 - x(x-1)(x-\lambda^{1/p}))^p.$$

The radical of the right-hand side gives the defining equation of X_p . Plugging in $\lambda^{1/p}$ gives $j(X_p) = j(X)^{1/p}$.

- (b) If X admits a degree p endomorphism $\pi : X \rightarrow X$, then $X \cong X_p$ by (2.5). Conversely, if $X_p \cong X$, then p_X factors as $F' \circ \hat{F}'$, where $F' : X_p \rightarrow X$ is the k -linear Frobenius morphism (Ex. 4.7). By (Ex. 4.15), the Hasse invariant of X is 0 if and only if $\hat{F}'$ is an inseparable morphism. The Frobenius is the unique purely inseparable morphism of degree p (2.5), so if $X_p \cong X$, then F' and $\hat{F}'$ must differ by an automorphism of X , i.e., they are associates in R . If F' and $\hat{F}'$ are associates, then $\hat{F}'$ is an inseparable morphism since F' is inseparable. Hence, the Hasse invariant of X is 0.

- (c) If $\text{Hasse}(X) = 0$, then $\hat{F}' : X \rightarrow X_p$ is a purely inseparable morphism of degree p . By (2.5), X and the p th Frobenius twist $(X_p)_p \cong X_{p^2}$ of X_p are isomorphic as abstract k -schemes. Hence, $j(X) = j(X)^{1/p^2}$ by (Ex. 4.20a).
- (d) Clearly f and F commute on the underlying topological spaces. The statement holds for any homomorphism between k -algebras, so it is immediate that f and F commute at the level of sheaves as well. If $\text{Hasse}(X) \neq 0$, then the p -linear map of cohomology $F^* : H^1(X, \mathcal{O}_X) \rightarrow H^1(X, \mathcal{O}_X)$ is non-zero, and commutes with $f^* : H^1(X, \mathcal{O}_X) \rightarrow H^1(X, \mathcal{O}_X)$. In particular, $F^*(1) \neq 0$, and $(f \circ F)^*(1) = (F \circ f)^*(1)$. Expanding both sides gives

$$\begin{aligned} (f \circ F)^*(1) &= F^*(f^*(1)) = F^*(\lambda_f \cdot 1) = \lambda_f^p \cdot F^*(1) \\ (F \circ f)^*(1) &= f^*(F^*(1)) = \lambda_f \cdot F^*(1). \end{aligned}$$

Since $F^*(1)$ is non-zero, we have $\lambda_f^p = \lambda_f$. Hence, λ_f is in $\mathbb{F}_p$.

□